

A Project Report On

Workout Tracker

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award of the degree

BACHELOR OF COMPUTER APPLICATIONS

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For subject

05BC3504-Data Visualisation

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DECLARATION

We hereby declare that this project work entitled Workout Tracker is a record done by us.

We also declare that the matter embodied in this project is genuine work done by us and has not been submitted whether to this University or to any other University for the fulfillment of any course of study.

Place: Marwadi University

Date: 25/08/2025

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1. Introduction

1.1 Overview of the Project

This project focuses on analyzing **workout and fitness data** using three different platforms: **Python (Matplotlib), Excel, and Tableau**.

Each platform is used to generate a different set of charts and visualizations to explore workout trends, user performance, and fitness insights.

The purpose is to compare the strengths of each tool and demonstrate how they complement each other for data-driven decision making.

1.2 Objective of the Analysis

- To apply multiple data visualization techniques on workout data.
- To compare three tools (Python, Excel, Tableau) for their effectiveness in exploring insights.
- To identify workout patterns (duration, calories, heart rate) across users.
- To create a **consolidated reporting framework** that combines static reports, coded analysis, and interactive dashboards.

1.3 Scope and Importance of Identifying Patterns and Trends in Data

Identifying workout trends helps in:

- Tracking calorie burn and exercise consistency.
- Comparing users' workout behaviors (duration, intensity, category).
- Understanding the relationship between **heart rate, duration, and calories**.
- Helping trainers and individuals optimize workout plans.

1.4 Dataset Overview

- **Type of Data:** Tabular (workout sessions per user).
- **Key Attributes:** User, Workout Type, Duration, Calories, Heart Rate, Difficulty, Date.
- **Nature:** Mixed (categorical + numerical + time series).
- **Purpose:** To analyze trends in fitness activity and compare visualization techniques across tools.

2. Data Collection

2.1 Dataset Description

- **Users:** Multiple users performing workouts.
- **Attributes:** Duration (minutes), Calories, Heart Rate, Workout Category (Cardio/Strength), Difficulty, Date.

2.2 Source of the Data

The dataset was simulated for academic purposes, inspired by real-world fitness tracking applications.

2.3 Data Collection Methodology

Data was structured to reflect realistic patterns of fitness activity. Each record represents one workout session by a user.

2.4 Data Format and Structure

- Format: CSV / Excel file
- Structure: Tabular, with rows = workouts, columns = attributes

2.5 Initial Data Exploration

- Checked for missing values.
- Verified numerical ranges (e.g., heart rate, calories).
- Categorized workouts into **Cardio** and **Strength** for comparisons.

3. Data Analysis and Data Visualization

3.1 Problem Statements, Solutions & Visualizations

(A) Python (Matplotlib) – 5 Charts

1. Scatter Plot – Calories vs Duration
2. Box Plot – Heart Rate by Category
3. Histogram – Workout Duration Frequency
4. Line Plot – Average Calories per Day
5. Bar Chart – Total Workouts per User

(B) Excel – 5 Charts

1. Stacked Bar Chart – Cardio vs Strength Workouts per User
2. Horizontal Bar Chart – Top Workouts by Average Calories
3. Histogram – Calories Burned Frequency
4. Box Plot – Duration Distribution per User
5. Grouped Bar Chart – Average Heart Rate by Category per User

(C) Tableau – 5 Charts

1. Pie Chart – Share of Calories Burned by User
2. Scatter Plot – Difficulty vs Calories
3. Line + Marker Chart – Heart Rate Trend per User
4. Area Chart – Total Calories Burned Over Time
5. Bubble Chart – Calories vs Duration (Bubble Size = Heart Rate)

4. Conclusion

4.1 Summary of Findings

- **Python:** Best for reproducible, code-based analysis.
- **Excel:** Best for static summaries & pivot-style comparisons.
- **Tableau:** Best for interactive dashboards and live exploration.

4.2 Implications of the Analysis

- Trainers can identify users who need performance improvement.
- Calories and duration correlation shows efficiency of workouts.
- Heart rate monitoring reveals workout intensity levels.

4.3 Limitations of the Project

- Dataset is small and simulated (not from live sensors).
- Analysis depends on accuracy of recorded data.
- Tableau requires a licensed tool for full functionality.

4.4 Future Work and Recommendations

- Expand dataset with **real fitness tracker data**.
- Apply **machine learning** to predict calories based on duration & difficulty.
- Develop a **combined dashboard** integrating Python, Excel, and Tableau outputs.

5. References

5.1 Data Sources

- Simulated fitness dataset (created for academic analysis).

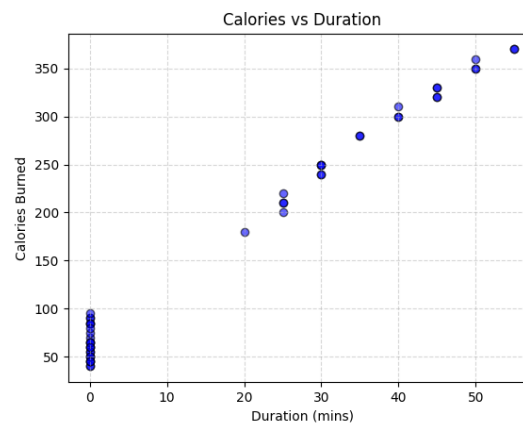
5.2 Tools and Libraries Used

- **Python:** Matplotlib
- **Excel:** Pivot Tables, Charts
- **Tableau:** Public Desktop Edition

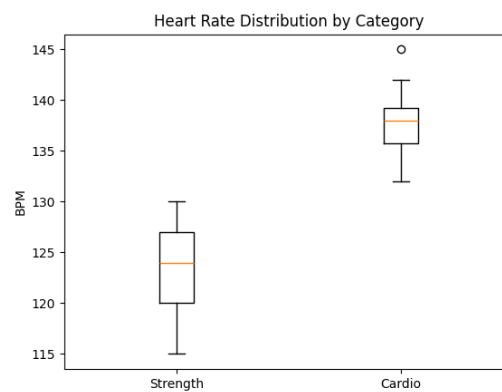
Figures

1) Python

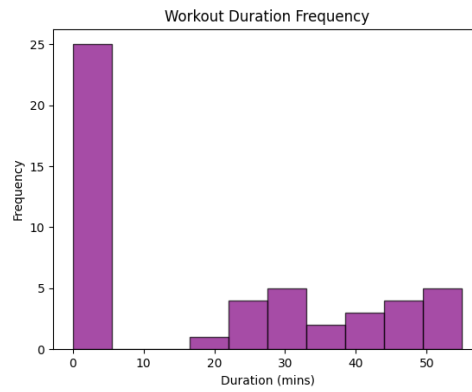
1. Scatter Plot – Calories vs Duration



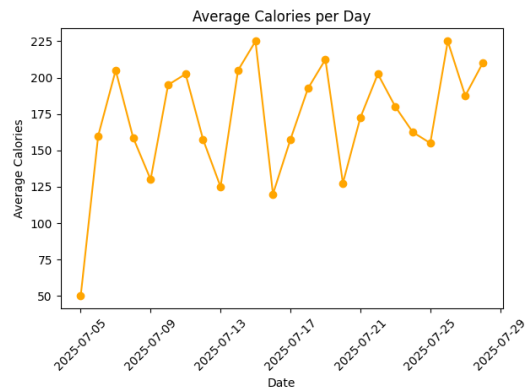
2. Box Plot – Heart Rate by Category



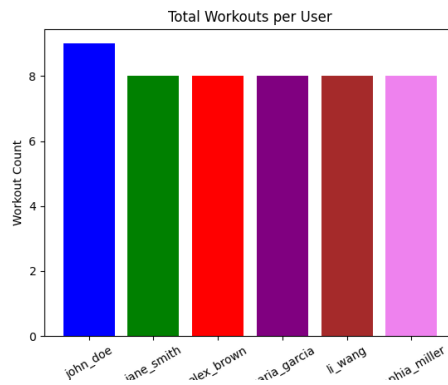
3. Histogram – Workout Duration Frequency



4. Line Plot – Average Calories per Day

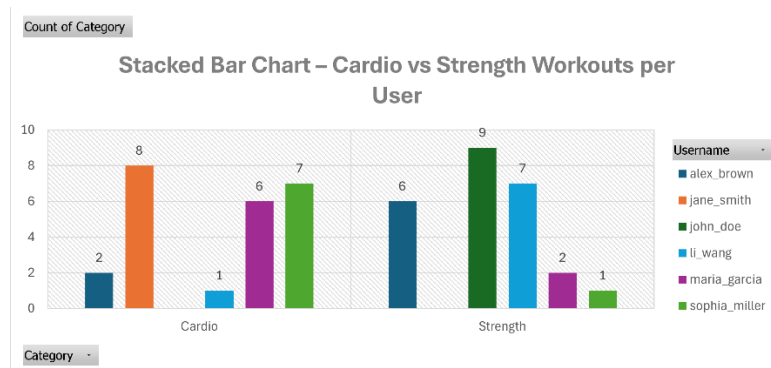


5. Bar Chart – Total Workouts per User

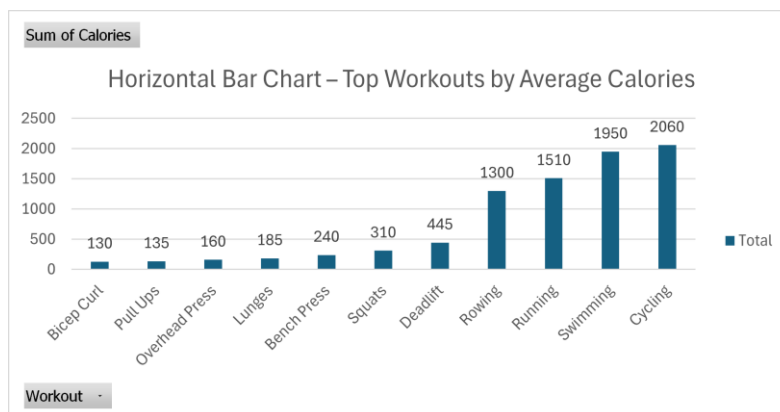


2) Excel

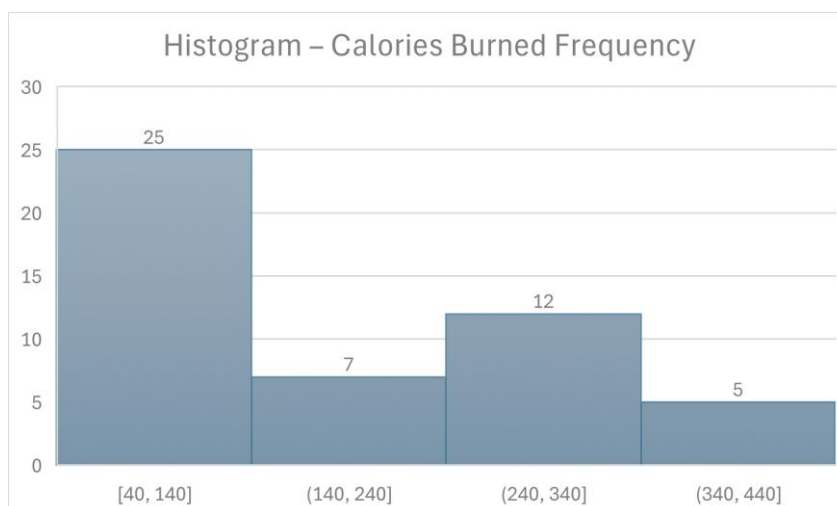
1. Stacked Bar Chart – Cardio vs Strength Workouts per User



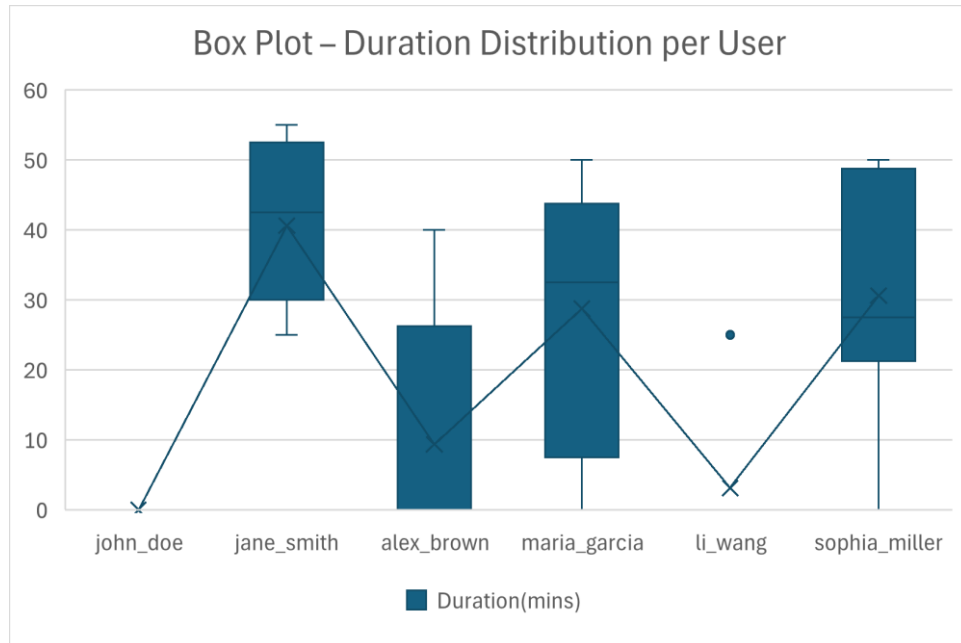
2. Horizontal Bar Chart – Top Workouts by Average Calories



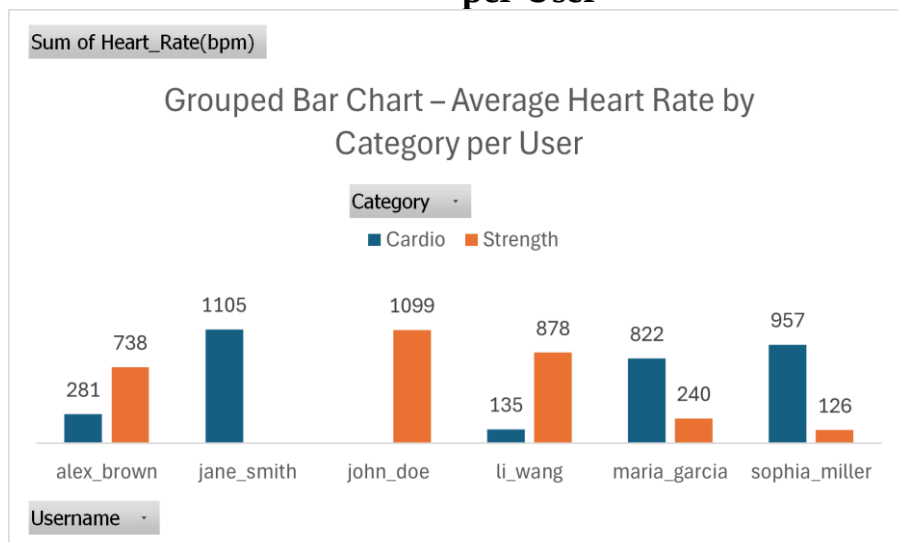
3. Histogram – Calories Burned Frequency



4. Box Plot – Duration Distribution per User

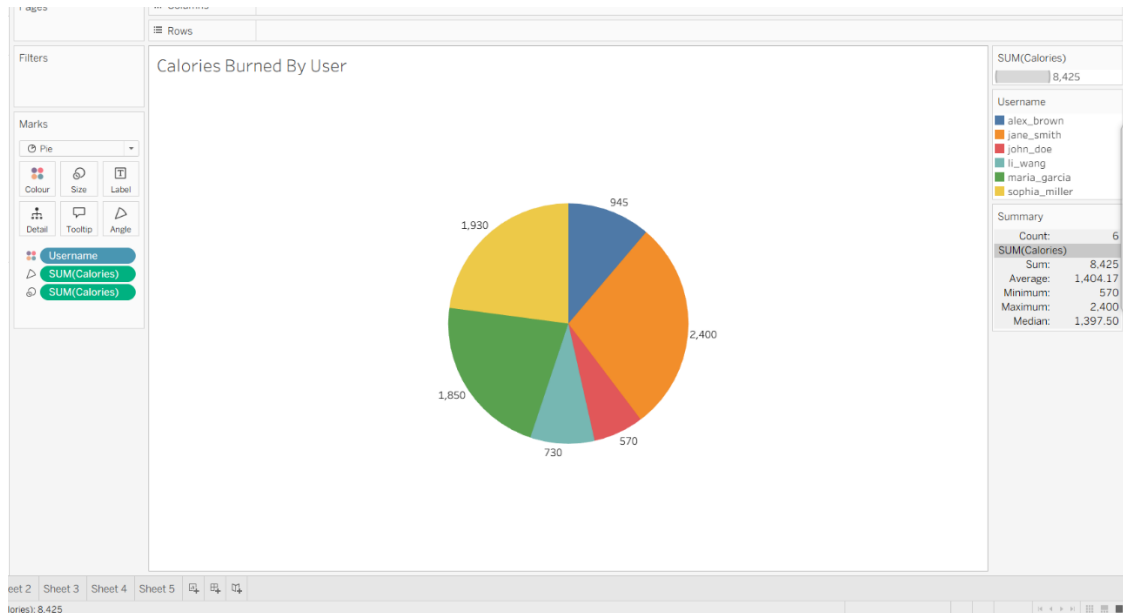


5. Grouped Bar Chart – Average Heart Rate by Category per User

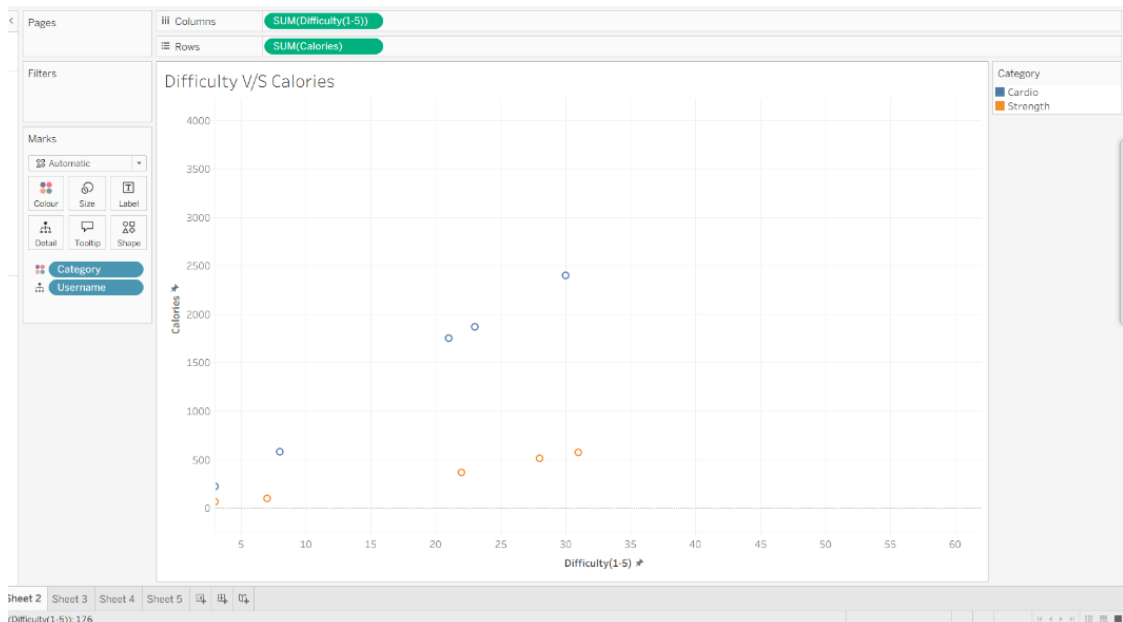


3) Tableau

1. Pie Chart – Share of Calories Burned by User



2. Scatter Plot – Difficulty vs Calories



3. Line + Marker Chart– Heart Rate Trend per User



4. Area Chart – Total Calories Burned Over Time



5. Bubble Chart – Calories vs Duration (Bubble Size = Heart Rate)

